Example:-

	x_1	x_2	y
1	0	0	
2	1	0	
3	1	0	
4	2	1	
5	3	1	

$\theta_1 = 0, \quad \theta_2 = 0$

$\alpha = 0.1$

After iteration 1 what is θ_1 & θ_2

Write:-

$$\frac{\partial}{\partial \theta_1} l(\theta) = \sum_{i=1}^{5} \left([y^{(i)} - h_\theta(x^{(i)})] \right.$$

$\theta^T x_1$ ~~$h_\theta(x^{(1)})$~~ : ~~$h_\theta(x^{(1)})$~~ $= 0 \times 1 + 0 \times 0 = 0$

$\theta^T x_2$ ~~$h_\theta(2)$~~ $= 0 \times 2 + 0 \times 1 = 0$

$\theta^T x_3$ ~~$h_\theta(3)$~~ $= 0 \times 3 + 0 \times 1 = 0$

$\theta^T x_4$ ~~$h_\theta(4)$~~ $= 0 \times 4 + 0 \times 2 = 0$

$\theta^T x_5$ ~~$h_\theta(5)$~~ $= 0 \times 5 + 0 \times 3 = 0$

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$$h_\theta(x^{(1)}) = \frac{1}{1+e^{-0}} \quad \frac{1}{2}$$

$$h_\theta(x^{(2)}) = \frac{1}{2}$$

$$h_\theta(x^{(3)}) = \frac{1}{2}$$

$$h_\theta(x^{(4)}) = \frac{1}{2}$$

$$h_\theta(x^{(5)}) = \frac{1}{2}$$

$$y^{(1)} - h_\theta(x^1) = 0 - \frac{1}{2} = -\frac{1}{2} \times 1$$

$$y^{(2)} - h_\theta(x^2) = 0 - \frac{1}{2} = -\frac{1}{2} \times 2 = -1$$

$$y^{(3)} - h_\theta(x^3) = -\frac{1}{2} \times 3 = -\frac{3}{2} = -1.5$$

$$y^{(4)} - h_\theta(x^4) = 1 - \frac{1}{2} = 0.5 \times 4 = 2$$

$$y^{(5)} - h_\theta(x^5) = 0.5 \times 5 = 2.5$$

$$\sum_{i=1}^{n} = 1.5$$

$$\theta_1 := \theta_1 + \alpha(1.5)$$

$$= 0 + 0.1 \times (1.5) = 0.15$$

θ_2 :-

$$\sum_{i=1}^{n} = \left(\frac{-1}{2} \times 0\right) + \left(\frac{-1}{2} \times 1\right) + \left(\frac{-1}{2} \times 1\right)$$

$$+ \left(1 - \frac{1}{2}\right) \times 2$$

$-1.5 + 1$

$$+ \left(1 - \frac{1}{2}\right) \times 3$$

$$= 0 - 0.5 - 0.5 + 1 + 1.5$$

$$= 1.5$$

$$\theta_2 = 0 + \alpha (1.5)$$

$$= 0 + 0.1 \times 1.5$$

$$= 0.15$$